

Algorithms Problem Solving

Level 1

ATM PIN Code



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Definition

- ◆ This program asks the user to enter an ATM PIN code and checks whether it matches the correct PIN.
- ◆ If the PIN is `1234`, the program displays the account balance.
- ◆ If the PIN is incorrect, the program displays an error message and asks for the PIN again.
- ◆ The program uses strings, a `do-while` loop, a Boolean condition, and functions.

Explanation

- ◆ ``ReadPinCode()`` reads the PIN as a ``string``.
- ◆ Reading the PIN as a string is useful because a PIN is an identifier, not a number used in mathematical calculations.
- ◆ ``LogIn()`` repeatedly asks for the PIN until the correct PIN is entered.
- ◆ If the PIN is ``"1234"``, ``LogIn()`` returns ``true``.
- ◆ If the PIN is incorrect, it prints ``Wrong PIN Code`` and changes the console color.
- ◆ ``CheckAndShowBalance()`` also reads the PIN and checks it directly.
- ◆ When the PIN is correct, it changes the console color and displays the balance.
- ◆ In ``main()``, the program calls:
``CheckAndShowBalance(7500);``

Steps in main()

1. Read the PIN code as a string.
2. Compare the entered PIN with `"1234"`.
3. If the PIN is correct, display the balance.
4. If the PIN is incorrect, display `Wrong PIN Code`.
5. Ask the user to enter the PIN again.
6. Repeat until the correct PIN is entered.

💡 Example

◆ If the user enters:

1111

◆ The PIN is incorrect.

◆ The program asks again:

1234

◆ The PIN matches the correct PIN.

➡ Result

Your account balance is 7500

Note

- ◆ The PIN is stored and compared as a string:
`PinCode == "1234"`
- ◆ This is useful for PIN codes because leading zeros can be preserved. For example, `"0123"` would remain four characters.
- ◆ The program uses `do-while` because the user must enter the PIN at least once.

⊘ Warning

- ◆ The function ``LogIn()`` is defined but never called in ``main()`` or ``CheckAndShowBalance()``.
- ◆ Therefore, the actual program uses the PIN-checking logic inside ``CheckAndShowBalance()`` and ``LogIn()`` is currently unnecessary.
- ◆ ``system("color ...")`` is Windows-specific and is not portable C++.



Comparison

▨ Correct PIN

- ✓ Entered PIN equals `"1234"`.
- ✓ The loop ends.
- ✓ The balance is displayed.
- ✓ Console color is changed to indicate success.

▨ Incorrect PIN

- ✓ Entered PIN does not equal `"1234"`.
- ✓ ``Wrong PIN Code`` is displayed.
- ✓ The user is asked to enter the PIN again.
- ✓ The balance is not displayed.

✦ Summary

- ✓ PIN codes can be stored as strings.
- ✓ `do-while` can repeatedly request input until a valid value is entered.
- ✓ A correct PIN allows the balance to be displayed.
- ✓ An incorrect PIN causes the program to ask again.
- ✓ `LogIn()` is currently unused in the program.

Thank you 😊